

Q The use of Wireless Multi-Channel Audio Systems (WMAS) technology marks a significant shift in wireless audio transmission. How does Sennheiser Spectera's bidirectional communication and wideband approach improve spectral efficiency and reduce interference compared to traditional narrowband systems?

A Spectera leverages a single wideband RF channel, either 6 or 8 MHz depending on local regulations, to transmit bidirectional audio and control data. This contrasts with traditional systems that rely on multiple unidirectional narrowband carrier frequencies of 200 kHz bandwidth. When a source of interference hits this narrowband channel or when RF fading occurs, the microphone or in-ear will be disturbed. As Spectera uses the entire wideband channel of 8 or 6 MHz, it can perfectly work around this source of interference.

With regard to spectral efficiency, we've already seen that IEMs and mics can simply be put in the same TV channel, which makes workflow a lot easier. Also, the operator can freely use the wideband RF channel, and decide whether more links are needed, and whether latency should be lowest, or whether range needs to be highest because of a special show element. These factors influence each other, and can easily be adapted to the local situation.

Also, all components within the Sennheiser Spectera ecosystem permanently sense the spectrum. For the first time, it is possible to see what is happening on your own transmission frequency and take informed action when interference occurs.

Q With Sennheiser Spectera offering full remote control, monitoring, and permanent spectrum sensing, how do you envision this system transforming the workflow for professionals managing complex audio setups in large-scale productions like festivals and broadcast events?



A The new workflow becomes apparent when you no longer need to calculate complex frequency charts, or cable up a tower of rack-mount mic receivers and in-ear transmitters. With Spectera, you choose a centre frequency, and the system will organise its audio links in the RF channel.

Spectera enables professionals to control and monitor audio settings, RF health, and battery status remotely via a continuous data stream. This simplifies adjustments during performances or broadcasts without the need for direct physical interaction with the mobile equipment. Also, engineers can constantly monitor the RF environment and detect interference before it becomes problematic.

At large festivals, there's another advantage that Spectera can play out: While the transmitted power adds up with narrowband systems, Sennheiser Spectera has a low spectral density. This makes it easier to reuse frequencies, also between neighbouring theatres, or in a broadcasting complex.

And the ecosystem will definitely make a difference when it comes to touring productions. As said, you have 32 inputs and 32 outputs in a 1U, 19" unit – this means you could carry your multi-channel system in a trolley if needed – and not in several flight cases.

Q Sennheiser has a rich legacy of wireless innovation. How does

Spectera's development process, from its conception to global regulatory approval, compare to previous milestones in Sennheiser's history, such as the introduction of the first wireless microphone system?

A Just as Sennheiser's first wireless microphone system in 1957 represented a transformative leap from wired to wireless audio, Spectera signifies a major evolution from single-direction narrowband systems to bidirectional wideband transmission. This transition underscores Sennheiser's commitment to pushing the boundaries of what is possible, focusing on advancements such as bidirectional communication, remote control capabilities, and spectrum sensing.

The Spectera ecosystem reflects the company's dedication to staying at the forefront of technological innovation, much like it did with its initial wireless breakthroughs. Spectera introduces a level of modularity that allows the ecosystem to be updated and expanded, ensuring its relevance and adaptability to future industry demands.

Spectera stands out as a monumental development in Sennheiser's illustrious history, addressing modern challenges with completely new technology. Sennheiser Spectera sets a new standard for professional audio set-ups in both live and studio environments, continuing Sennheiser's tradition of transforming the landscape of wireless audio. **d**